Bank Guards as a Coordination Game — class version

Lecture 5 case. Maheshri & Mastrobuoni (2021), *REStat* 103(3), 547–562. Short version of `deterrence-game.pdf`, where every step is derived. **[Paper]** = the paper; **[Ours]** = our analysis. Costs in €; payoff = −cost.

1. The model as a game [Paper] (pp. 548–550) with our payoffs [Ours]

Banks $i = 1, \dots, N$ choose $g_i \in \{0, 1\}$ (guard or not). A guard costs c_i; a robbery costs L_i; the robbery probability is $p(g_i, g_{-i})$. Robbers are implicit: displacement is their best response to visible guards. Payoff $u_i = -(p\,L_i + c_i g_i)$. Bank i hires iff

$$\Delta(g_{-i}) = p(0, g_{-i}) - p(1, g_{-i}) \;>\; \theta_i = c_i / L_i \qquad \text{(eq. 5)}.$$

Displacement falls on unguarded banks, so Δ *rises* with the others' guards: **strategic complements**, hence a coordination game. Order banks $\theta_1 \le \cdots \le \theta_N$; every equilibrium is "the first e banks guard" (Lemma, Prop. 1, p. 549).

Calibration (Table 3 col. 7, pp. 555–556; base risk 0.07): $p(g, x) = 0.07 - 0.0318\,g + 0.0169\,x - 0.0336\,g\,x$, with x the share of neighbours guarded. So $\Delta(0) = 0.0318$, $\Delta(1) = 0.0654$, and 0.531 of the deterrence effect is displaced.

2. Two banks, $c = 40{,}000$ [Ours]

Best responses: guard against an unguarded neighbour iff $L > c/\Delta(0) = 1{,}257{,}862$; against a guarded one iff $L > c/\Delta(1) = 611{,}621$. Below 611,621 the unique equilibrium is NN; above 1,257,862 it is GG; in between both are equilibria. Costs in € thousand (row, column), $L = 1{,}000{,}000$ ($\theta = 0.04$):

	G	N
G	**61.5**, **61.5**	78.2, 86.9
N	86.9, 78.2	**70.0**, **70.0**

Pure equilibria GG and NN; mixed: each guards with probability 0.244, which is worse for both banks (expected cost 74,124 euros). A **stag hunt**: GG is Pareto better (123.0 against 140.0 in total) but NN is self-confirming: "too few guards". At $L = 700{,}000$ ($\theta = 0.0571$) the costs are NN 49.0, GG 55.05, mixed $\Pr(G) = 0.754$ and GG is the Pareto-dominated equilibrium: "too many guards" (**[Paper]** Prop. 2, p. 550). Cutoffs: GG is Pareto better iff $\theta < 0.0485$ ($L > 824{,}742$) and risk dominant iff $\theta < 0.0486$; they nearly coincide because $\beta_2 \approx -\beta_3/2$.

Discrepancy. The paper's 1.37M and 662k (p. 557) reproduce with $c = 44{,}000$ and Table 4 col. 2, not with the stated 40,000 (which gives 1,257,862 and 611,621; footnote 24 mentions 44,000).

3. Four banks [Ours]

$\theta = (0.020, 0.044, 0.050, 0.068)$, $c = 40{,}000$, so $\Delta(k/3) = 0.0318,\ 0.0430,\ 0.0542,\ 0.0654$ for $k = 0, \dots, 3$ guarded others. Enumerating all 16 profiles: pure equilibria $\{1\}$ (total cost 290,154) and $\{1, 2, 3\}$ (271,510). The social optimum, all four (252,393), is *not* an equilibrium: bank 4 pays 52,647 but only 51,118 if it fires (gain 1,529). A ban (nobody) costs 300,813. A requirement gives 252,393. A subsidy $s = 1{,}529.4$ makes bank 4 indifferent; $s = 1{,}600$ gives the unique equilibrium all four. **[Paper]** Price instruments are ill suited (p. 550); here the subsidy works only because θ_4 is close to $\Delta(1)$. A Figure 1-style market $\theta = (0.025, 0.035, 0.058, 0.060)$ has equilibria $\{1, 2\}$ and all four.

4. Lecture 5: the repeated game [Ours]

The 2000–2009 panel is a finitely repeated game, $T = 10$. **Any sequence of stage equilibria is an SPE:** a one-stage deviation changes only the stage payoff, and the stage payoff cannot rise because the profile is a stage NE (one-stage deviation principle).

Carrot and stick, two periods. Play all four in $t = 1$; in $t = 2$ play $\{1\}$ if bank 4 complied (its cost 44,490: the carrot is *fewer guards around the unguarded bank*) and $\{1, 2, 3\}$ if it fired (its cost 51,118: the stick, itself a stage NE). Bank 4 complies iff

$$52{,}647 + \delta \cdot 44{,}490 \le 51{,}118 + \delta \cdot 51{,}118 \iff \delta \ge \frac{1{,}529}{6{,}628} = 0.2307.$$

Banks 1–3 do not want to fire at $t = 1$ for any δ. **Nash reversion fails forever:** both stage NE are cheaper for bank 4 (44,490 and 51,118) than all four (52,647), so deviating always pays. Hence quantity regulation (a requirement), or a coordination device such as the Italian Banking Association (**[Ours]** suggestion), beats relying on repetition.

Evidence [Paper] (pp. 552–553, Figs. 3–5): neighbours increasingly make the same guard decision (over 40% of pairs with a guard have both guarded in 2000, against about 20% by chance), and firings outnumber hirings.

5. Observability, in one line

Guards are visible (p. 548), which is what lets robbers displace; unobservable devices (LoJack, Ayres–Levitt) give a positive externality instead. Link to Lecture 4: *observability matters*.

Takeaway

- Visible guards displace crime onto unguarded banks, so guarding decisions are strategic complements: a stag hunt with two equilibria, either of which can be the efficient one.
- Whether policy should promote or restrict guards is therefore ambiguous (Prop. 2); a requirement or ban forces a profile, while a subsidy tips the market only when θ's sit near the thresholds.
- Repetition (carrot and stick) can sustain all four for two periods when $\delta \geq 0.2307$, but Nash reversion cannot sustain it forever.